

Data, Dataset and Their Types

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Data and Datasets

- **What is Data?**
- Data refers to raw facts, measurements, observations, symbols, or values collected from real-world entities, systems, or processes.
- **Examples** include: student's GPA, temperature measured by a sensor

- **What is Dataset?**
- A dataset is a collection of related data objects organized for storage, processing, and analysis.
- A dataset may contain: records, transactions, sequences, graphs, images, videos, embeddings
- Unlike individual data values, datasets emphasize:
  - structure
  - organization
  - relationships
  - temporal dynamics
  - spatial dependencies
  - …

Datasets Examples

- Examples:
  - **sales dataset**: customers, store items, sales
  - **medical dataset**: patients, treatments
  - **university dataset**: students, professors, courses
- Database rows -> data objects
- Database columns ->attributes

Important Characteristics of Datasets

- **Datasets** are used for analysis and can vary significantly in **Size, Dimensionality**, **Sparsity**, **Resolution** and **Distribution**.

- **Dimensionality** (number of attributes in a dataset)
  - High dimensional data can lead to challenges
  - Challenges such as the **curse of dimensionality**.
- **Sparsity**
  - Proportion of zero or missing values in a dataset
  - Only presence counts
- **Resolution**
  - Level of detail represented in the dataset
  - Patterns depend on the scale
- **Size**
  - Amount of data, often measured in bytes or number of records.
  - Type of analysis may depend on size of dataset
- **Distribution**
  - The way values are spread across the dataset
  - Such as centrality and dispersion

Customer dataset example: Collection of Customers Records

ID	Age	Experience	Income	ZIP Code	Family	CCAvg	Educ.	Mortgage	Personal Loan	Securities Account
1	25	1	49	91107	4	1.60	1	0	0	1
4	35	9	100	94112	1	2.70	2	0	0	0
5	35	8	45	91330	4	1.00	2	0	0	0
6	37	13	29	92121	4	0.40	2	155	0	0
9	35	10	81	90089	3	0.60	2	104	0	0
11	65	39	105	94710	4	2.40	3	0	0	0
12	29	5	45	90277	3	0.10	2	0	0	0
18	42	18	81	94305	4	2.40	1	0	0	0
20	55	28	21	94720	1	0.50	2	0	0	1
23	29	5	62	90277	1	1.20	1	260	0	0
26	43	19	29	94305	3	0.50	1	97	0	0
27	40	16	83	95064	4	0.20	3	0	0	0
29	56	30	48	94539	1	2.20	3	0	0	0
31	59	35	35	93106	1	1.20	3	122	0	0
32	40	16	29	94117	1	2.00	2	0	0	0
35	31	5	50	94035	4	1.80	3	0	0	0
36	48	24	81	92647	3	0.70	1	0	0	0
37	59	35	121	94720	1	2.90	1	0	0	0
38	51	25	71	95814	1	1.40	3	198	0	0
40	38	13	80	94115	4	0.70	3	285	0	0
41	57	32	84	92672	3	1.60	3	0	0	1

Customer Dataset Example

OBS#	CHK_ACCT	DURATION	HISTORY	NEW_CAR	USED_CAR	FURNITURE	RADIO/TV	EDUCATION	RETRAINING	AMOUNT	SAV_ACCT	RESPONSE
1	0	6	4	0	0	0	1	0	0	1169	4	1
8	1	36	2	0	1	0	0	0	0	6948	0	1
16	0	24	2	0	0	0	1	0	0	1282	1	0
24	1	12	4	0	1	0	0	0	0	1804	1	1
32	0	24	2	0	0	1	0	0	0	4020	0	1
40	1	9	2	0	0	0	1	0	0	458	0	1
48	0	6	2	0	1	0	0	0	0	1352	2	1
56	3	6	1	1	0	0	0	0	0	783	4	1
64	1	48	0	0	0	0	0	0	1	14421	0	0
72	3	7	4	0	0	0	1	0	0	730	4	1
80	1	30	2	0	0	1	0	0	0	3832	0	1
88	1	36	2	0	0	0	0	1	0	12612	1	0
96	1	54	0	0	0	0	0	0	1	15945	0	0
104	1	9	4	0	0	1	0	0	0	1919	0	1
112	2	15	2	0	0	0	0	1	0	392	0	1

Types of Data vs Types of Datasets

- Types of data describe **the nature of individual values or attributes**.
- Examples include:
  - **numerical data**
  - **categorical data**
  - **ordinal data**
  - **binary data**
  - **text data**
  - **temporal data**
  - image data
  - spatial coordinates
  - More complex data

Types of Datasets

- Types of datasets describe **how collections of data are organized and related**.
- Examples include:
  - **tabular datasets**
  - **transactional datasets**
  - **matrix datasets**
  - time-series datasets
  - graph and network datasets
  - stream datasets
  - spatial datasets
  - multimedia datasets
  - embedding and vector datasets
- This distinction becomes increasingly important in intelligent systems because learning algorithms frequently depend more on dataset structure than on individual attribute types.

Example Types of Datasets

- **Tabular datasets**:
  - rows represent data objects
  - columns represent attributes
  - most common dataset structures in traditional analytics.
- **Examples** include: student databases, sales records, hospital records
- Traditional machine learning algorithms are highly effective for tabular datasets.
  - decision trees
  - random forests
  - XGBoost
  - logistic regression

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- **Transactional datasets**: special forms of record data where each record contains a set of items.
- **Example** transaction: {milk, bread, eggs, coffee}
- Applications include: association rule mining, recommendation systems, retail analytics, consumer behavior analysis

Basic Types of Data (or Attributes)

- **Attributes (**or **columns**, **dimensions**, **features**, **variables**): a data field, representing a characteristic or feature of a data object.
  - Example: *name, address, etc.*
- Attributes can be classified as:
- **Categorical**
  - **Nominal:**
    - **Categories without a meaningful order** (e.g., gender, colors, marital status).
    - Can be used to measure similarity or dissimilarity
  - **Ordinal:**
    - Categories with a **meaningful order** but no fixed distance between categories.
    - Can be used to measure similarity or dissimilarity, and compare (e.g., rankings, grade, size).
  - **Binary:** Two categories (e.g., true/false, yes/no).
- **Numerical**
  - **Discrete:** Countable values (e.g., number of students).
  - **Continuous:** Measurable quantities that can take any value within a range (e.g., height, weight).

Attributes Types

- **Categorical Data**: qualitative variables having discrete values
  - **Nominal Data**: color, marital-status
  - **Ordinal Data**: grade, product-size
  - **Binary Data**: gender, yes/no
- **Numerical Data**: quantitative variables

Student_ID	Gender	Prev_Exam_Marks	Height (cm)	Weight Caregory (kgs)	Play Cricket
S001	M	65	178	61	1
S002	F	75	174	56	0
S003	M	45	163	62	1
S004	M	57	175	70	0
S005	F	59	162	67	0

Categorical Attributes

- **Nominal:** categories, states, or "names of things"
  - *Hair_color* = {*auburn, black, blond, brown, grey, red, white*}
  - marital status, occupation, ID numbers, zip codes
- **Ordinal**
  - Values have a meaningful order (ranking) but magnitude between successive values is not known.
  - *Size* = {*small, medium, large*}, grades, army rankings
- **Binary**
  - Nominal attribute with only 2 states (0 and 1)
  - Symmetric binary: both outcomes equally important
    - e.g., gender
  - Asymmetric binary: outcomes not equally important.
    - e.g., medical test (positive vs. negative)
    - Convention: assign 1 to most important outcome (e.g., HIV positive)

Discrete vs. Continuous Attributes

- **Discrete Attribute**
  - Has only a **finite** set of values
    - E.g., zip codes, profession, or the set of words in a collection of documents
  - Sometimes, represented as integer variables
- **Continuous Attribute**
  - Has real numbers as attribute values
    - E.g., temperature, height, or weight
  - Practically, real values can only be measured and represented using a finite number of digits
  - Continuous attributes are typically represented as floating-point variables.

Attributes Types

	Attribute Type	Description	Examples	Operations
Categorical Qualitative	Nominal	Nominal attribute values only distinguish. (=, □)	zip codes, employee ID numbers, eye color, sex: {*male, female*}	mode, entropy, contingency correlation, $\chi 2$ test
	Ordinal	Ordinal attribute values also order objects. (<, >)	hardness of minerals, {*good, better, best*}, grades, street numbers	median, percentiles, rank correlation, …
Numeric Quantitative	Interval	For interval attributes, differences between values are meaningful. (+, -)	No have true zero such as calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, *t* and *F* tests
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	Have true zero such as weight, monetary quantities, counts, age, mass, length,	geometric mean, harmonic mean, percent variation

This categorization of attributes is due to S. S. Stevens

Roles of Attributes

- **Target / Response / Outcome / Dependent variable/Label (output): PlayCricket**
- **Predictor / Independent (input): Gender, Prev_Exam_Marks, Height, Weight**

Student_ID	Gender	Prev_Exam_Marks	Height (cm)	Weight Caregory (kgs)	Play Cricket
S001	M	65	178	61	1
S002	F	75	174	56	0
S003	M	45	163	62	1
S004	M	57	175	70	0
S005	F	59	162	67	0

Data Sources

- **Data sources** are categorized into two main types: Primary Data and Secondary.

- **Primary Data:** Raw data collected directly by the data analysis team from the source.
  - **Examples**: data from applications, surveys, experiments, direct interviews, or data collected from sensors.
  - **Advantages:** The data is accurate and tailored to the specific needs of the data analysis team.
  - **Disadvantages:** Collecting this type of data requires significant time and expense.

- **Secondary Data:** Data that the analysis team does not collect themselves but obtain from other existing sources.
  - **Examples:** internet, private organizations, research institutes, books, journals, government agencies
  - **Advantages:** This data is readily available, saving time and costs associated with data collection.
  - **Disadvantages:** The data may not fully meet the specific needs of the analysis team because it was not collected for their purposes.

Similarity, Dissimilarity, Proximity

- Similarity
  - Numerical measure of how alike two data objects are
  - **Higher similarity values indicate objects are more alike**
  - Often falls in the range [0,1]
- Dissimilarity (e.g., distance)
  - Numerical measure of how different two data objects are
  - **Higher dissimilarity values indicate greater differences**
  - Lower when objects are more alike
  - Minimum dissimilarity is often 0
- Proximity
  - Refers to **both similarity** and **dissimilarity**
  - Indicating the "closeness" or "distance" between objects

Similarity and Dissimilarity between Attributes

- **Nominal variables**
  - Similarity

$$Similarity\ (a, b) = 1\ if\ a = b$$
$$Similarity\ (a, b) = 0\ if\ a \neq b$$

  - Dissimilarity

$$Dissimilarity\ (a, b) = 0\ if\ a = b$$
$$Dissimilarity(a, b) = 1\ if\ a \neq b$$

- **Ordinal variables**
  - Similarity

$$Similarity\ (a, b) = 1\ - |\ ord(a)\ - ord(b)|\ /(n - 1)$$

  - Dissimilarity

$$Dissimilarity\ (a, b) = |\ ord(a)\ - ord(b)|\ /(n - 1)$$

  - Where ord(a) and ord(b) are integer order between 0 and n.

- **Numerical variables**
  - Dissimilarity

$$Dissimilarity\ (a, b) = |\ a\ - b|$$

  - Similarity

$$Similarity\ (a, b) = -\ |\ a\ - b|$$

Similarity and Dissimilarity between Attributes

Nominal Attributes

If we have two variables a and b representing colors, where $a = \text{"red"}$ and $b = \text{"red"}$, then:

- $\text{Similarity}(a, b) = 1$ (because both values are the same).
- $\text{Dissimilarity}(a, b) = 0$ (because both values are the same).

Ordinal Attributes

Consider variables a and b representing sizes: "small", "medium", and "large". We can assign:

- "small" = 0
- "medium" = 1
- "large" = 2

So, if $a = \text{"small"}$ and $b = \text{"large"}$:

- $\text{ord}(a) = 0$
- $\text{ord}(b) = 2$
- $n = 3$

$$\text{Similarity}(a, b) = 1 - \frac{|0 - 2|}{3 - 1} = 1 - \frac{2}{2} = 1 - 1 = 0$$

Numerical Attributes

If $a = 10$ and $b = 15$:

$$\text{Dissimilarity}(a, b) = |10 - 15| = 5$$

$$\text{Similarity}(a, b) = -|10 - 15| = -5$$

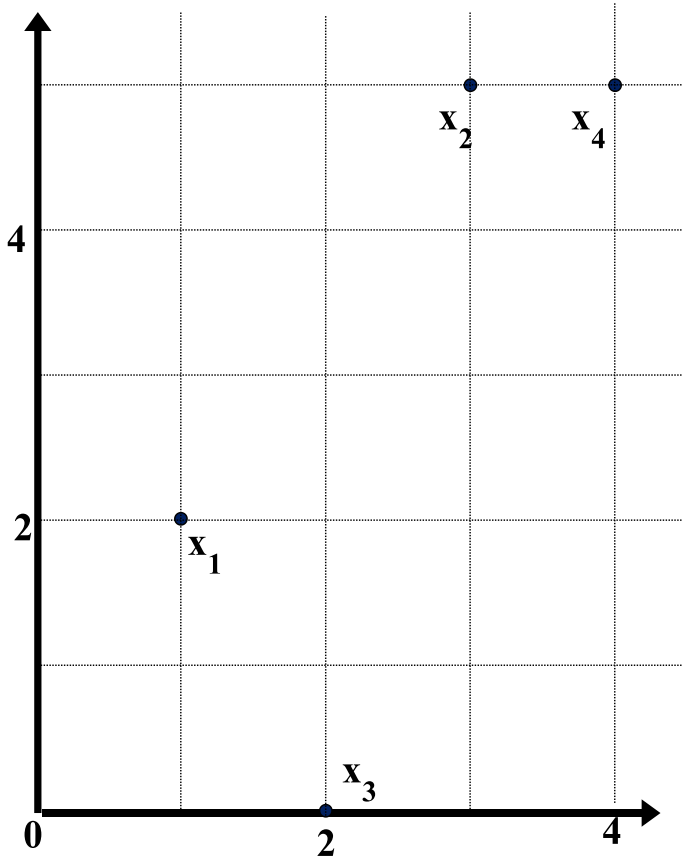
Dissimilarity (Distance) between Numerical Data

Euclidean Distance is most popular:

$$d_{ij} = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \cdots + (x_{ip} - x_{jp})^2}$$

- **Euclidean Distance** measures the straight-line distance between two points in a multidimensional space
- **Problem**: Raw distance measures are highly influenced by scale of measurements
- **Solution**: Normalization

Example:
Data Matrix and Dissimilarity Matrix



Data Matrix

point	attribute1	attribute2
x1	1	2
x2	3	5
x3	2	0
x4	4	5

Dissimilarity Matrix

(with Euclidean Distance)

	x1	*x2*	*x3*	*x4*
x1	0			
x2	3.61	0		
x3	5.1	5.1	0	
x4	4.24	1	5.39	0

Dissimilarity (Distance) between Numerical Data

- *Minkowski distance*: A popular distance measure, a generalization of Euclidean distance.

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \cdots + |x_{ip} - x_{jp}|^h}$$

where $i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \ldots, x_{jp})$ are two p-dimensional data objects, and h is the order (the distance so defined is also called L-h norm)

- Properties
  - d(i, j) > 0 if i ≠ j, and d(i, i) = 0 (Positive definiteness)
  - d(i, j) = d(j, i) (Symmetry)
  - d(i, j) ≤ d(i, k) + d(k, j) (Triangle Inequality)
- A distance that satisfies these properties is a metric

Special Cases of Minkowski Distance

- *h* = 1: Manhattan (city block, L_1 norm) distance
  - E.g., the Hamming distance: the number of bits that are different between two binary vectors

$$d(i,j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \ldots + |x_{ip} - x_{jp}|$$

- *h* = 2: (L_2 norm) Euclidean distance

$$d(i,j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \ldots + |x_{ip} - x_{jp}|^2)}$$

Dissimilarity Measures for Nominal Data

- Method 1: Simple matching
  - *m*: # of matches, *p*: total # of attributes

$$d(i,j) = \frac{p - m}{p}$$

- Method 2: Use a large number of binary attributes
  - creating a new binary attribute for each of the *M* nominal categories

Dissimilarity Measures for Binary Data

- A contingency table for binary data
- q=(1,1)
- r=(1,0)
- s=(0,1)
- t=(0,0)

Object *i* \ Object *j*	1	0	sum
1	q	r	$q+r$
0	s	t	$s+t$
sum	$q+s$	$r+t$	p

- Distance measure for symmetric binary variables:

$$d(i,j) = \frac{r+s}{q+r+s+t}$$

- Distance measure for asymmetric binary variables:

$$d(i,j) = \frac{r+s}{q+r+s}$$

Dissimilarity Measures for Binary Data

- Example

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	N
Mary	F	Y	N	P	N	P	N
Jim	M	Y	P	N	N	N	N

  - Gender is a symmetric attribute
  - The remaining attributes are asymmetric binary
  - Let the values Y and P be 1, and the value N 0

$$d(jack, mary) = \frac{0+1}{2+0+1} = 0.33$$

$$d(jack, jim) = \frac{1+1}{1+1+1} = 0.67$$

$$d(jim, mary) = \frac{1+2}{1+1+2} = 0.75$$

Dissimilarity Measures for Binary Data

- Example

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	1	0	1	0	0	0
Mary	F	1	0	1	0	1	0
Jim	M	1	1	0	0	0	0

  - Gender is a symmetric attribute
  - The remaining attributes are asymmetric binary
  - Let the values Y and P be 1, and the value N 0

$$d(jack, mary) = \frac{0+1}{2+0+1} = 0.33$$

$$d(jack, jim) = \frac{1+1}{1+1+1} = 0.67$$

$$d(jim, mary) = \frac{1+2}{1+1+2} = 0.75$$

$$d(i, j) = \frac{r+s}{q+r+s}$$

  - q=(1,1)
  - r=(1,0)
  - s=(0,1)
  - t=(0,0)

Similarity Measures for Binary Data

- Jaccard coefficient (*similarity* measure for *asymmetric* binary variables):

Object *i* \ Object *j*	1	0	sum
1	q	r	$q+r$
0	s	t	$s+t$
sum	$q+s$	$r+t$	p

$$sim_{Jaccard}(i, j) = \frac{q}{q + r + s}$$

Cosine Similarity

- A **document** can be represented by thousands of attributes, each recording the *frequency* of a particular word (such as keywords) or phrase in the document.

Document	*team*	*coach*	*hockey*	*baseball*	*soccer*	*penalty*	*score*	*win*	*loss*	*season*
Document1	5	0	3	0	2	0	0	2	0	0
Document2	3	0	2	0	1	1	0	1	0	1
Document3	0	7	0	2	1	0	0	3	0	0
Document4	0	1	0	0	1	2	2	0	3	0

- Other vector objects: gene features in micro-arrays, …
- Applications: information retrieval, biologic taxonomy, gene feature mapping, ...
- **Cosine Similarity**: Measures the cosine of the angle between two vectors.
- **Cosine Similarity** : If d_1 and d_2 are two vectors (e.g., term-frequency vectors), then

$$\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| \, ||d_2|| ,$$

where $\bullet$ indicates vector dot product, $||d||$: the length of vector d

Example: Cosine Similarity

- $\cos(d_1, d_2) = (d_1 \bullet d_2) / ||d_1|| \, ||d_2||$,
  where $\bullet$ indicates vector dot product, $||d||$: the length of vector d

- Ex: Find the **similarity** between documents 1 and 2.

$d_1 =$ (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)
$d_2 =$ (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)

$d_1 \bullet d_2 = 5*3+0*0+3*2+0*0+2*1+0*1+0*1+2*1+0*0+0*1 = 25$
$||d_1|| = (5*5+0*0+3*3+0*0+2*2+0*0+0*0+2*2+0*0+0*0)^{0.5} = (42)^{0.5} = 6.481$
$||d_2|| = (3*3+0*0+2*2+0*0+1*1+1*1+0*0+1*1+0*0+1*1)^{0.5} = (17)^{0.5} = 4.12$
$\cos(d_1, d_2) = 0.94$

Other Similarity and Dissimilarity Measures

- Simple Matching Coefficients (SMC)
- Pearson's Correlation
- Mahalanobis Distance
- Coherence similarity
- Similarities between data with various types of attributes
- Similarities between **Complex Data**
- Etc.